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(54) **INVERSE VOLTAGE-TO-CURRENT
CONVERSION CIRCUIT**

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H03K 5/26 (2006.01)

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H03K 5/19; **G01R 19/255**; **Y02B 70/10**;
H02M 1/32; **H02M 3/156**; **H02M 3/1588**;
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See application file for complete search history.

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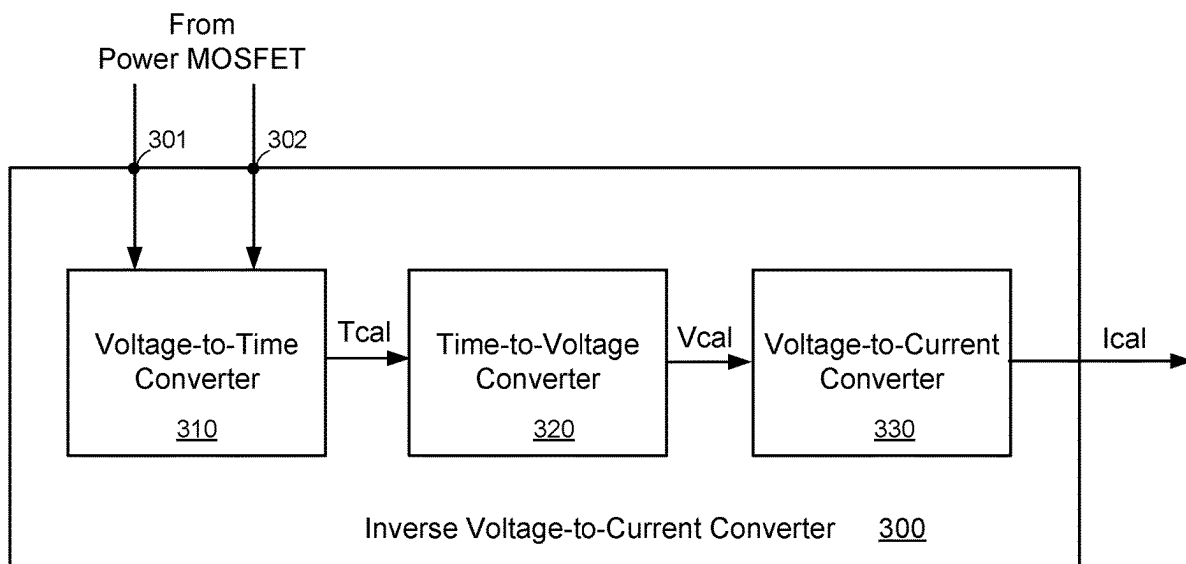
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(57) **ABSTRACT**

An inverse voltage-to-current conversion circuit for provid-
ing a current that is inversely related to an input voltage is
disclosed. A first input terminal and a second input terminal
receives the input voltage between the first and the second
input terminals. A voltage-to-time converter circuit provides
a time indicator pulse signal with a pulse width related to
inverse magnitude of the input voltage. A time-to-voltage
converter circuit provides a voltage indicator signal having
a magnitude based on the pulse width of the time indicator
pulse signal. A voltage-to-current converter circuit provides
a current indicator signal having a magnitude proportional to
the voltage indicator signal—the current indicator signal
being inversely related to the magnitude of the input voltage.

15 Claims, 8 Drawing Sheets



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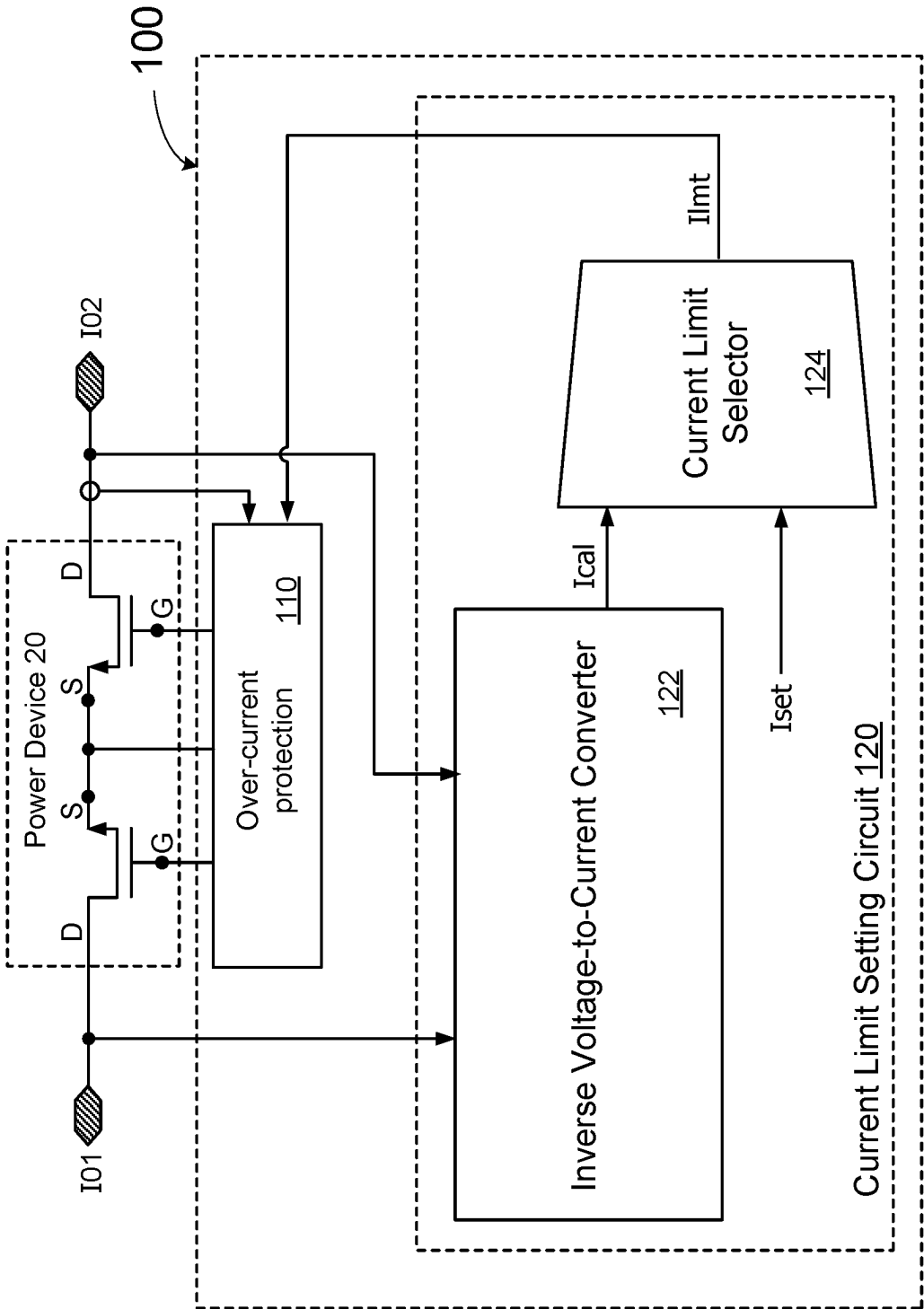


FIG. 1

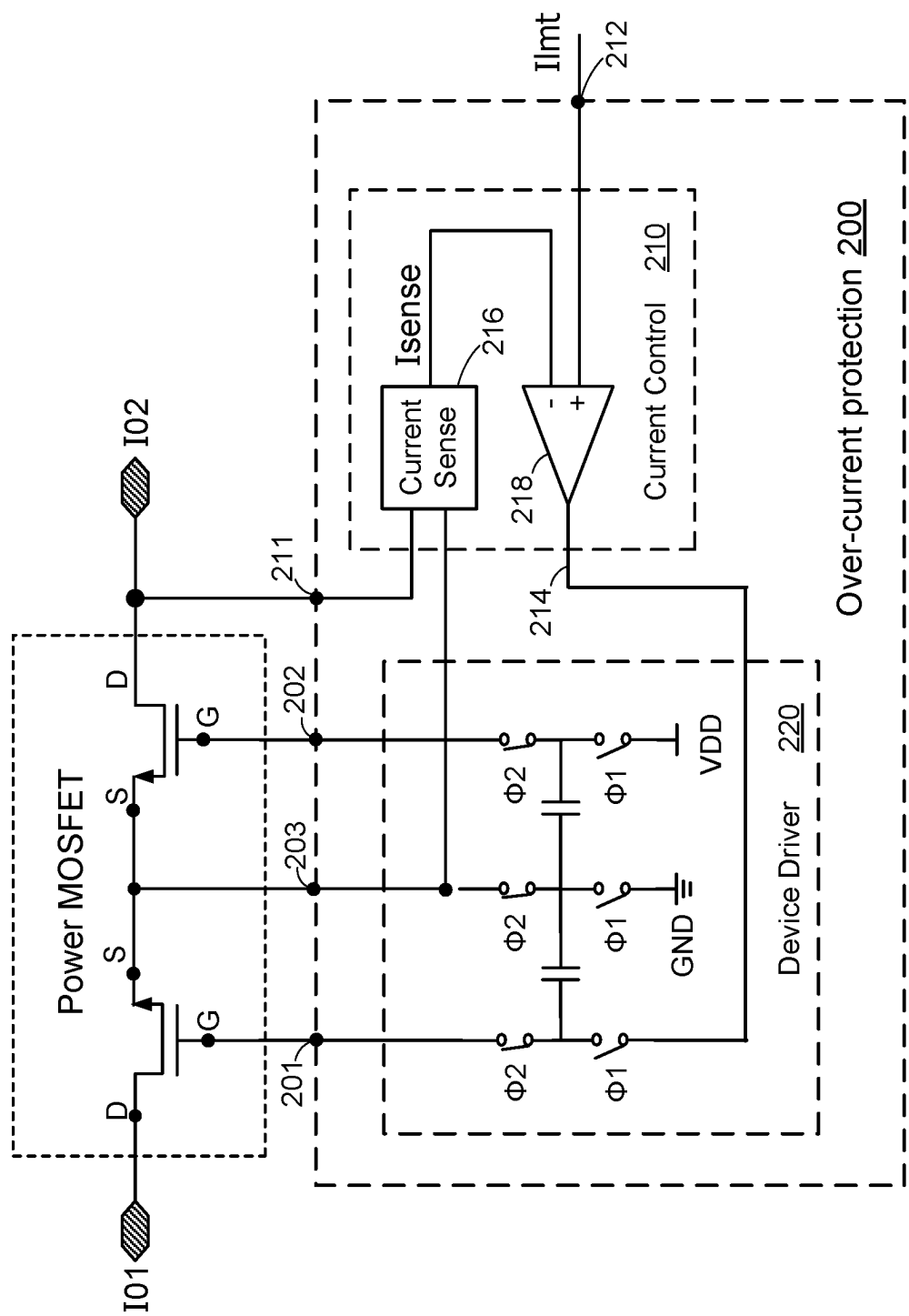


FIG. 2

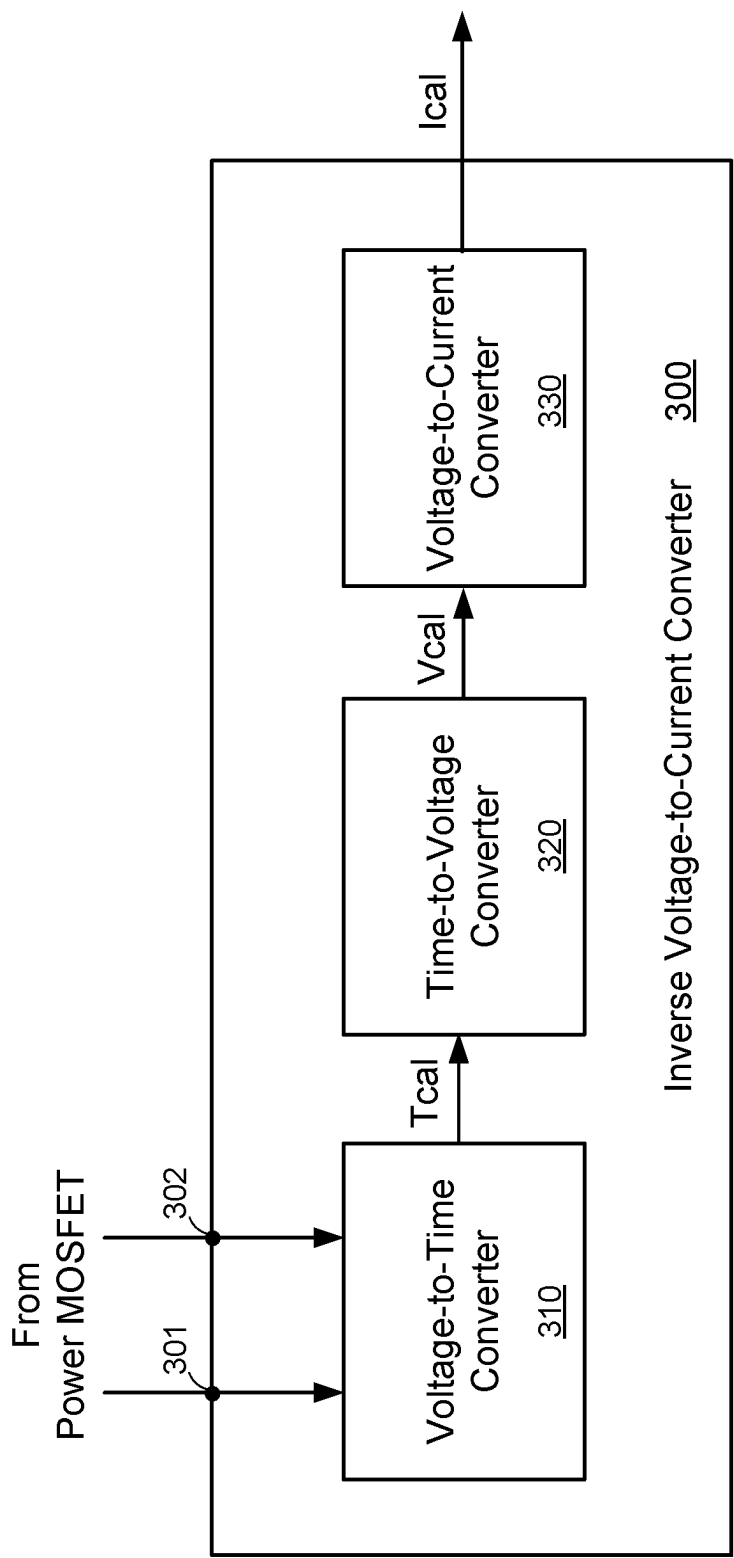


FIG. 3

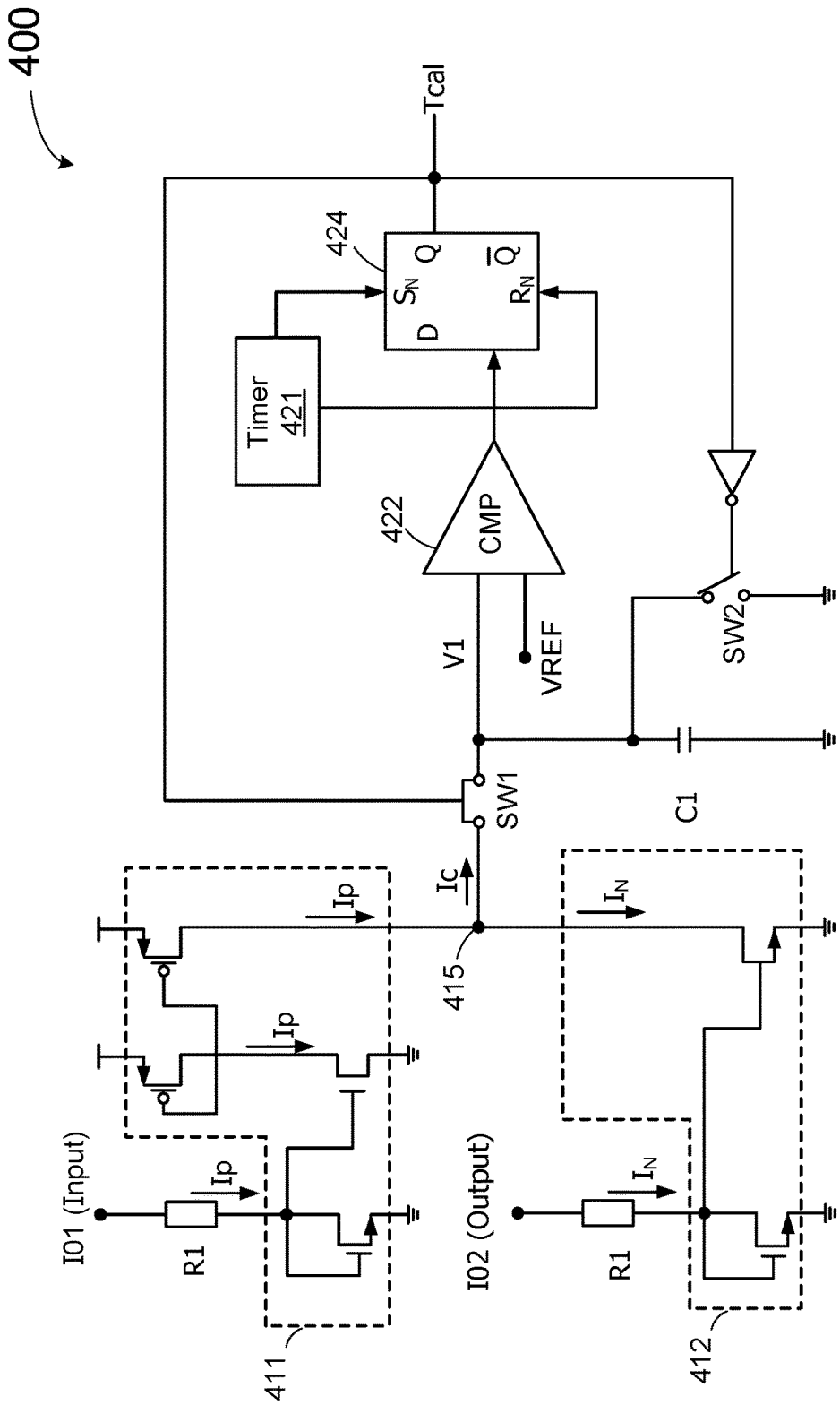
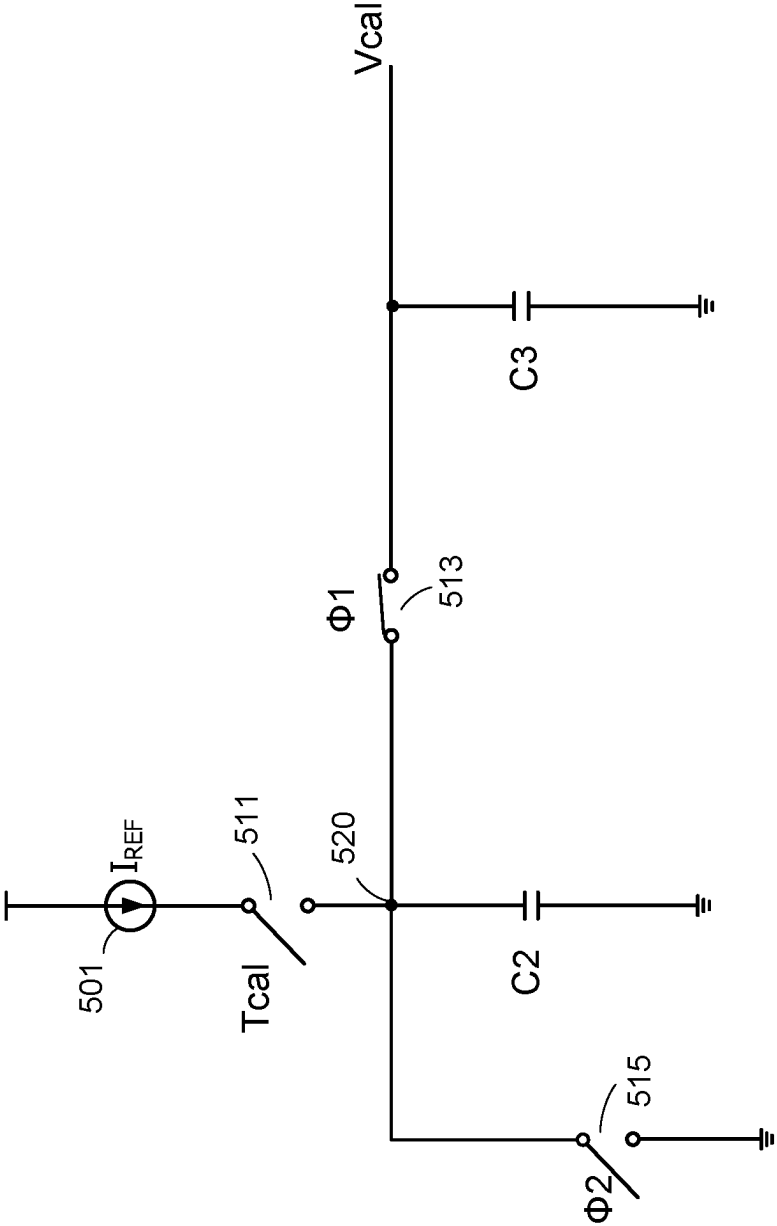


FIG. 4
Voltage-to-Time Converter

500



Time-to-Voltage Converter

FIG. 5

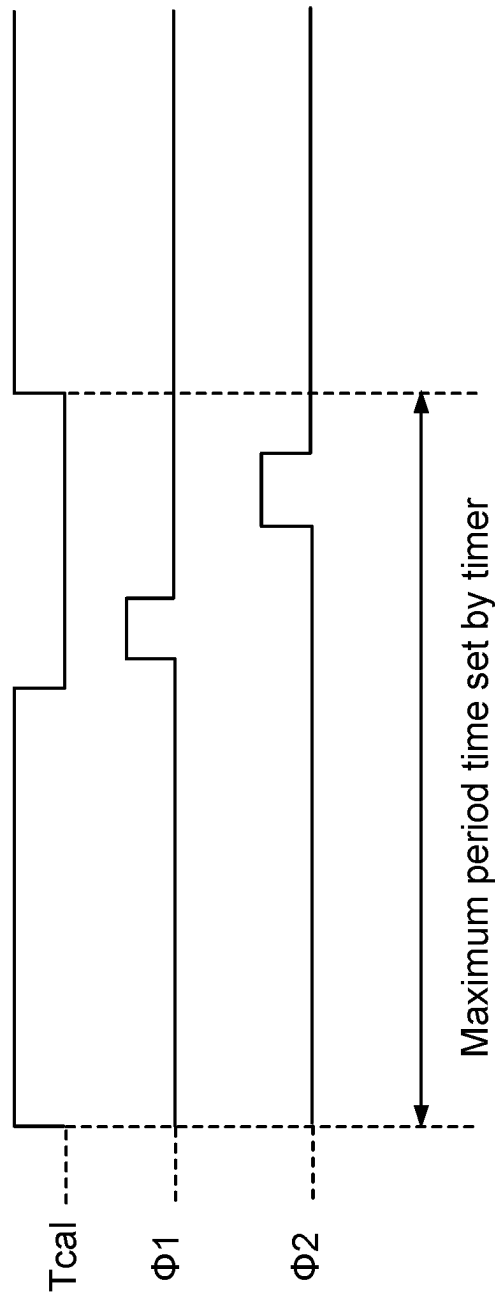


FIG. 6

700

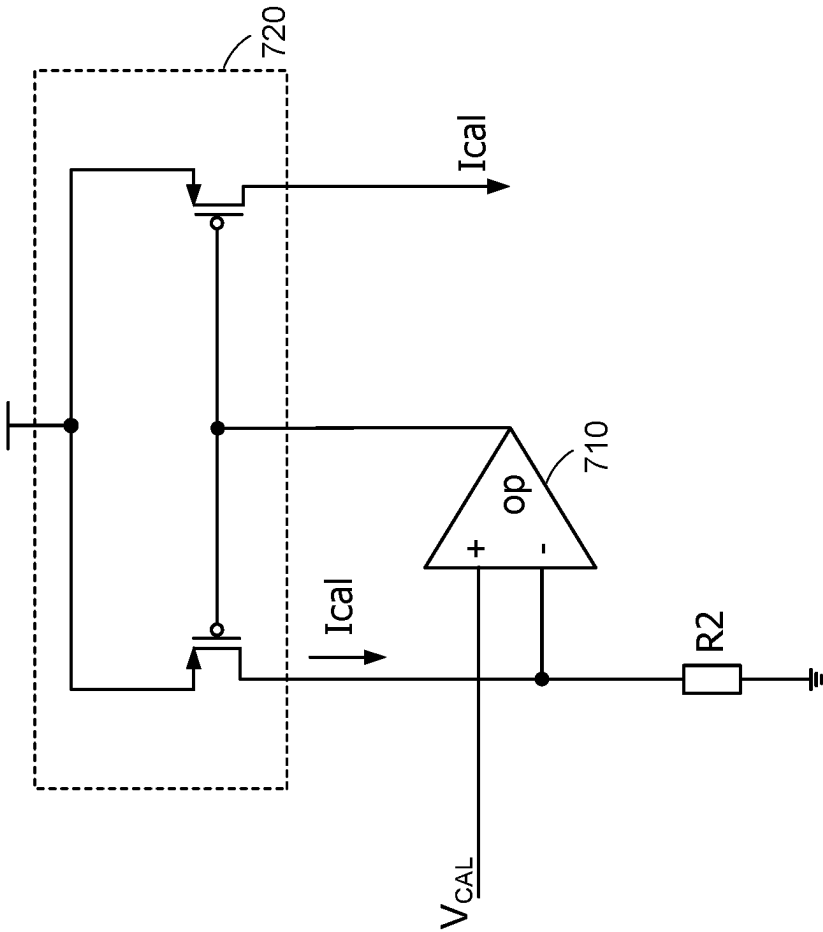
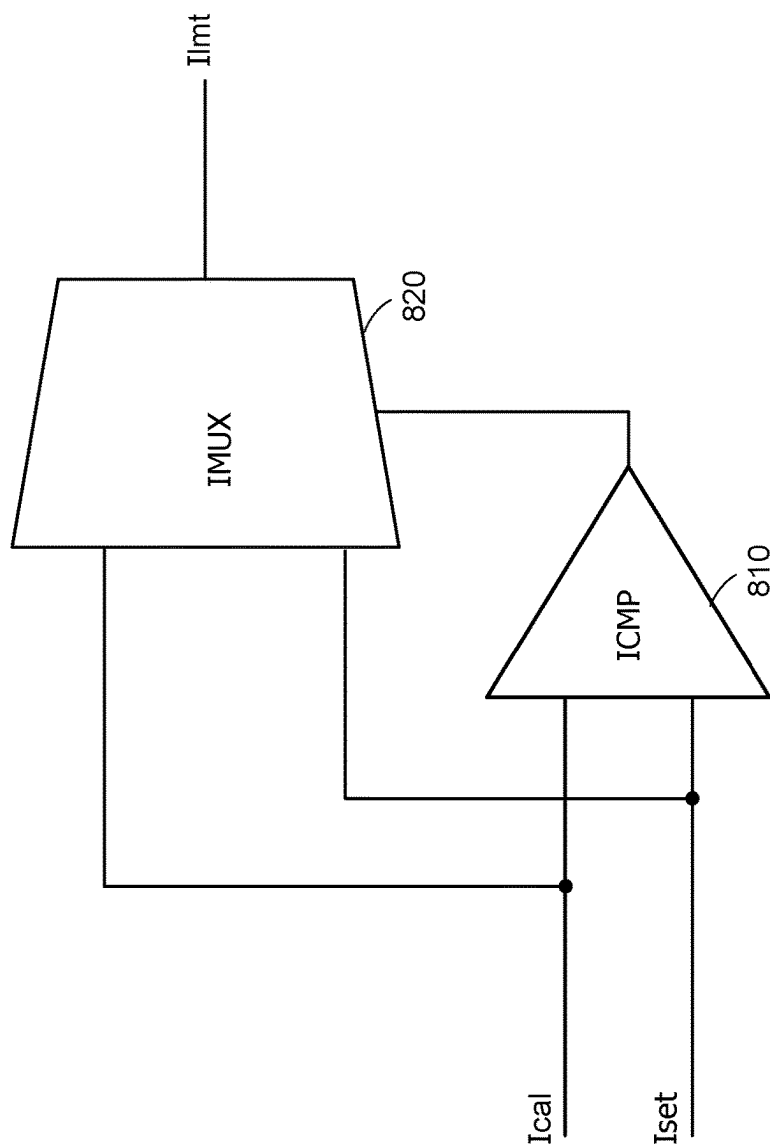


FIG. 7 Voltage-to-Current Converter

800



Current Selector

FIG. 8

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INVERSE VOLTAGE-TO-CURRENT CONVERSION CIRCUIT

CROSS REFERENCES TO RELATED APPLICATIONS

This application is a Divisional of and claims priority to U.S. patent application Ser. No. 16/046,991, filed on Jul. 26, 2018, which claims to U.S. Provisional Patent Application No. 62/537,954, filed on Jul. 27, 2017 and entitled "POWER SWITCH OVER-POWER PROTECTION" the disclosure of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

Embodiments of the invention are directed to integrated circuits. More particularly, the embodiments provide a system and method for over-current protection and over-power protection of a protected device.

In electronic circuits, over-current protection circuits are widely used. In an over-current protection circuit, a power device, e.g., a power MOSFET (Metal-Oxide-Semiconductor field effect transistor), is often used in a protection circuit to protect a load device against over-current conditions. The over-current protection circuit monitors the current flowing through the power device, and compares the measured current with a pre-set maximum allowable current to determine an over-current condition. In normal operations, the on-resistance R_{ON} of the power switch is kept low to allow current flow to the load device. When an over-current condition is detected, the protection circuit increases the on-resistance to reduce the voltage and current delivered to the load device.

The inventor has recognized a drawback with conventional over-current protection circuits. In an over-current event, the electrical power being consumed by the power device in the over current protection circuit can rise above the power limit. The situation is much worse in high voltage applications because in the same over current ratio (required output load current vs limited output load current), voltage at a high voltage (e.g., 20 V) can be several times as large than at a low voltage (e.g., 5 V). The power consumed on power switch can be several times larger as well, causing the power MOSFET more likely to operate beyond the specified condition in the SOA (Safer Operating Area). A Power MOSFET operating beyond its power limit can degrade its performance, reduce its life time, and can even be damaged. Therefore, conventional over-current protection circuits are not satisfactory in protection against over-power conditions.

BRIEF SUMMARY

Conventional over-current protection circuit usually has a pre-set current limit for the device being protected. However, there is often no provision to protect against over-power conditions. The pre-set current limit is often determined for safe device operation under normal operating voltages. As described above, when an over-current event happens, the on-resistance R_{on} of the power MOSFET is increased by the protection circuit in order to limit the current, and reduce the output voltage across a load. However, the voltage across the power MOSFET is raised because of the higher R_{on} . In another example, if the load device has been operating with a current below an over-current condition, and the load resistance drops, the voltage of the power MOSFET is increased. As a result, the power

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consumption can increase beyond what is expected from a device under normal operation. Power consumed on the power MOSFET is the product of current through it and voltage across it. The heavier the over current, the larger the voltage, then the higher the power consumption.

Embodiments of this invention teach an over-power protection circuit and method, in which the voltage across a power device, e.g., an MOSFET, is monitored, and a maximum current limit I_{CAL} is determined based on a pre-defined power limit threshold P_{LMT} . The maximum current limit I_{CAL} can be calculated by the power limit threshold P_{LMT} divided by the voltage across the power MOSFET. The power limit threshold P_{LMT} can be determined based on the SOA requirement. The maximum current limit I_{CAL} can vary with the voltage across the MOSFET. Therefore, a constant worst case current limit may be overly restrictive. The over-power protection device continuously monitors the voltage across the MOSFET, and dynamically determines a suitable maximum current limit I_{CAL} based on the monitored voltage across the device being protected to prevent over-power conditions.

An over-power protection circuit includes an over-current protection circuit that is configured to sense a current flowing through the device, compare the sensed current with a pre-set current limit, and prevent the current through the device from exceeding the current limit. The over-power protection circuit also includes a current limit setting circuit that is configured to provide a current limit I_{LMT} to the over-current protection circuit. The current limit can be either a pre-set current limit I_{SET} , which can be determined from the device specification, or a current limit I_{CAL} , based on the power limit of the device. As described above, the current limit I_{CAL} can be a maximum current allowed to go through power MOSFET that is calculated from a pre-defined power limit threshold P_{LMT} divided by the voltage across the power MOSFET.

According to some embodiments, a protection circuit includes an over-current protection circuit for coupling to a protected device, the protected device having a pre-set maximum current limit and a pre-set maximum power limit. The protection circuit also includes a current limit setting circuit coupled to the over-current protection circuit and the protected device, the current limit setting circuit configured to provide a target current limit signal to the over-current protection circuit for limiting current through the protected device. The target current limit signal is the lower one of the pre-set maximum current limit, and a current indicator signal determined based on the pre-set maximum power limit and a voltages across the protected device.

In an embodiment of the above protection circuit, values for circuit components in the over-current and over-power protection circuit are selected so that the current indicator signal represents a maximum current allowed based on the pre-set maximum power limit for a measured voltage across the protective device.

According to some embodiments, an over-power protection circuit for a MOSFET includes an over-current protection circuit and a current limit setting circuit, and an over-power protection circuit configured to continuously monitor a voltage across the MOSFET being protected to prevent over-power conditions, and to dynamically determine a maximum current limit based on the monitored voltage and a pre-set maximum power limit.

In some embodiments, the over-current protection circuit is configured to sense a current flowing through a protected device; compare the sensed current with a target current limit; and limit the current through the protected device to

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below the target current limit. The current limit setting circuit is configured to provide the target current limit to the over-current protection circuit, wherein the target current limit is either a pre-set current limit from a device specification, or a second current limit based on a pre-defined power limit.

According to some embodiments of the invention, an inverse voltage-to-current conversion circuit is provided for producing a current that is inversely related to an input voltage. The circuit includes a first input terminal and a second input terminal for receiving the input voltage between the first and the second input terminals. The circuit further includes a voltage-to-time converter circuit for providing a time indicator pulse signal with a pulse width related to inverse magnitude of the input voltage, and a time-to-voltage converter circuit for providing a voltage indicator signal having a magnitude based on the pulse width of the time indicator pulse signal. The circuit also includes a voltage-to-current converter circuit for providing a current indicator signal having a magnitude proportional to the voltage indicator signal, the current indicator signal being inversely related to the magnitude of the input voltage.

According to some embodiments of the invention, a method is provided for producing a current that is inversely proportional to a first voltage. The method includes sensing a first voltage across a first terminal and a second terminal, forming a voltage pulse signal with a pulse width inversely related to a magnitude of the first voltage, forming a second voltage having a magnitude based on a length of the voltage pulse signal, and forming a current signal having a magnitude proportional to the second voltage. The current signal is configured to have a magnitude inversely related to the first voltage.

Definitions

The terms used in this disclosure generally have their ordinary meanings in the art within the context of the invention. Certain terms are discussed below to provide additional guidance to the practitioners regarding the description of the invention. It will be appreciated that the same thing may be said in more than one way. Consequently, alternative language and synonyms may be used.

A power switch as used herein refers to a semiconductor switch, for example, a transistor, that is designed to handle high power levels.

A power MOSFET is a specific type of metal oxide semiconductor field-effect transistor (MOSFET) designed to handle significant power levels. An example of a power MOSFET for switching operations is called double-diffused MOS or simply DMOS.

A regulator or voltage regulator is a device for automatically maintaining a constant voltage level.

A constant-current regulator is a regulator that provides a constant output current. A constant current or constant voltage is understood to be a current or voltage that maintains a constant value with a range of deviation depending on design and manufacturing process variations or within a limitation according to a specification, for example, within $\pm 10\%$, $\pm 5\%$, or $\pm 1\%$.

An operational amplifier (op-amp or opamp) refers to a DC-coupled high-gain electronic voltage amplifier with a differential input and, usually, a single-ended output. An operational amplifier can be characterized by a high input impedance and a low output impedance, and can be used to perform mathematical operations in analog circuits.

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A voltage reference or a reference voltage source is an electronic device that ideally produces a fixed (constant) voltage irrespective of the loading on the device, power supply variations, temperature changes, and the passage of time.

A reference voltage is a voltage value that is used as a target for a comparison operation.

A current reference or a reference current source is an electronic device that ideally produces a fixed (constant) current irrespective of the loading on the device, power supply variations, temperature changes, and the passage of time.

A reference current is a current value that is used as a target for a comparison operation.

A sensed signal is a voltage or current signal determined by a sensing circuit.

When the term “the same” is used to describe two quantities, it means that the values of two quantities are determined the same within measurement limitations.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified schematic diagram illustrating an over-power protection circuit for protecting a power device that embodies certain aspects of this invention;

FIG. 2 is a simplified schematic diagram illustrating an over-current protection circuit that embodies certain aspects of this invention;

FIG. 3 is a simplified schematic diagram illustrating a voltage-to-current converter circuit for providing an output current I_{CAL} that is inversely related to an input voltage that embodies certain aspects of this invention;

FIG. 4 is a simplified schematic diagram illustrating a voltage-to-time converter circuit for providing a voltage pulse signal with a pulse width inversely related to a magnitude of an input voltage that embodies certain aspects of this invention;

FIG. 5 is a simplified schematic diagram for a time-to-voltage converter circuit that embodies certain aspects of this invention;

FIG. 6 is a waveform diagram illustrating the waveforms of T_{CAL} , Φ_1 , and Φ_2 described above in connection to FIG. 5 that embodies certain aspects of this invention;

FIG. 7 is a simplified schematic diagram illustrating a voltage-to-current converter circuit that embodies certain aspects of this invention; and

FIG. 8 is a simplified schematic diagram of a current selector circuit that embodies certain aspects of this invention.

DETAILED DESCRIPTION

Electronics devices, such as power MOSFETs, are usually designed to operate within its Safe Operation Area (SOA), which defines, among other things, how long a power MOSFETs can operate with a certain current flowing through it under a certain voltage. In embodiments of the invention, circuits and methods are provided using an over-current protection circuit to also provide over-power protection, i.e., to limit the power consumed in the power MOSFET to within the pre-set maximum power limit to protect the power MOSFET from performance degradation, short life time, and damages.

In some embodiments, the voltage across a power switch is monitored, and a safe current limit is determined from a power limit. The power limit can be determined, for example, from the Safe Operation Area (SOA) of the device.

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Given a power limit, the higher the voltage across the device, the lower the allowable current.

FIG. 1 is a simplified schematic diagram illustrating an over-power protection circuit for protecting a power device that embodies certain aspects of this invention. A power device is often used as a switch between a power supply and a load device for controlling the power provided to the load device. For example, in FIG. 1, a power device **20** can have a first terminal **IO1** for coupling to a power supply and a second terminal **IO2** for coupling to a load device. The power device **20** can include a single power transistor or a combination of power transistors. The power device can be a bipolar transistor or an MOSFET. In the example of FIG. 1, power device **20** represents an MOS power device being protected by an over-power protection circuit **100**. In this particular example, power device **110** includes the combination of two power MOSFETs **111** and **112**, each having a drain (D), source (S), and gate (G) terminals. It is understood that the circuits and methods described here are applicable to any device that needs to be protected against over-power conditions. In the description below, the device being protected, such as power device **20**, is also referred to as the protected device.

As shown in in FIG. 1, the over-power protection circuit **100** includes an over-current protection circuit **110** that is configured to sense a current flowing through the power device **20**, compare the sensed current with a current limit, and prevent the current flowing through the power device from exceeding the current limit. The over-power protection circuit also includes a current limit setting circuit **120** that is configured to provide a current limit I_{LMT} to the over-current protection circuit **110**. The current limit can be either a pre-set current limit I_{SET} , which can be determined from the device specification, or a current limit I_{CAL} based on the power limit of the power device and a measured current flowing through the power device. As described above, the current limit I_{CAL} can be a maximum current allowed to flow through power device, which can be determined from a pre-set maximum power limit threshold P_{LMT} divided by the voltage crossing the power MOSFET. For example, the power limit threshold P_{LMT} can be determined based on its SOA requirement.

As shown in FIG. 1, the current limit setting circuit **120** can include an inverse voltage-to-current converter **122** and a current limit selector circuit **124**. The inverse voltage-to-current conversion circuit **122** is configured for sensing a voltage across two terminals of the protected device, e.g., power device **20**, and provide a current that is inversely related to the voltage across the two terminals of the protected device. As described above, given a power limit P_{LMT} , the maximum current allowed in the device is inversely proportional to the voltage across the device. The output current of the inverse voltage-to-current conversion circuit I_{CAL} is the maximum allowed current based on the sensed voltage. The current limit setting circuit **120** also includes a current selector circuit **124** for selecting a current limit I_{LMT} from either the I_{CAL} described above or a current I_{SET} , which is a pre-et current limit for the over-current protection circuit, which may be based on the current carrying capability of the device, and may not include considerations for over-power protection.

In embodiments of the invention, values for circuit components in the over-current and over-power protection circuit are selected so that the current indicator signal represents a maximum current allowed based on the pre-set maximum power limit for a measured voltage across the protective device. More details are described below.

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FIG. 2 is a simplified schematic diagram illustrating an over-current protection circuit that embodies certain aspects of this invention. Over-current protection circuit **200** in FIG. 2 is an example of circuits which can be used as the over-current protection circuit **110** in FIG. 1. The over-current protection circuit **200** includes a current control circuit **210** and a device driver circuit **220**. In can be seen that over-current protection circuit **200** has two output terminals **201** and **202** coupled to the gates of the MOSFETs and a third output terminal **203** coupled to the source terminals of the power MOSFET for driving the MOSFETS. The over-current protection circuit **200** also has a first input terminal **211** for current sensing and a second input terminal **212** for receiving a target current limit signal I_{LMT} .

The current control circuit **210** generates a current control signal in response to comparing a sensed current signal in the protected device with the target current limit signal. In FIG. 2 current control circuit **210** compares the current sensed the current I_{sense} in the power MOSFET with the current limit I_{LMT} from the inputs and generates current control signal **214**, which is used to modify the gate drive voltage according to the comparison result to change the bias of the MOSFET and the current and voltage of the power MOSFET in order to limit the current and power in a load device.

The current control circuit **210** includes a current sense circuit **214** to sense current through power MOSFET by monitor voltage cross the output power MOSFET near the **102** terminal. Since the R_{dson} of a power MOSFET is pre-defined and designed, the voltage across the output power MOSFET V_{ds} is equal to $I \times R_{on}$, the sensed current I_{sense} can be converted from V_{ds} by a voltage to current converter.

A conventional current sense circuit can also be used. As an example, a small sampling MOSFET much smaller than the power MOSFET, for example, by a ratio of 1000:1, can be biased with the same drain, source, and gate voltages as the power MOSFET and provides a sensed current that represents the current through the power MOSFET.

In FIG. 2, current control circuit **210** includes a comparator circuit **218** with input signals I_{sense} and I_{LMT} . I_{sense} is a sensed current signal that represents a current flowing through the power MOSFETS, and I_{LMT} is a target current limit signal that triggers a current protection action. The output of the current control circuit **210** is fed to device driver circuit **220** to reduce the current in the power MOSFET. In some embodiments, comparator circuit **215** can include a current comparator that receives two input current signals I_{sense} and I_{LMT} and produces a current control signal **214**, which can be a voltage signal having a low value and a high value for driving the gate of the MOSFET. For example, the high voltage value can be a power supply voltage V_{DD} , and a low voltage value can be a preset safe gate voltage to allow a safe current in the MOSFET.

The device driver circuit **220** can includes a charge-pump driver having two separate charge-pumps. The charge pump driver can include multiple capacitors and switches. The charge pump for input power MOSFET (connected to terminal **IO1**) is for current limit control, so the V_{gs} can be reduced and regulated when an over-current event occurs. The charge pump for output power MOSFET (connected to terminal **IO2**) is directly connected to a power supply V_{DD} , so that this MOSFET is fully turned on.

Charge pump gate driver illustrates the gate drive circuit for the power MOSFET close to the input terminal. When the sensed power MOSFET current I_{sense} reaches the

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current limit, the gate driver circuit switches the gate drive voltage to a preset safe gate voltage.

As shown in FIG. 2, in the first time interval, timing signal $\Phi 1$ closes two switches to charge the capacitor to a voltage, for example, VDD at 5V, when there is no over-current or over-power condition. When the over-current or over-power condition is met, signal $\Phi 2$ closes corresponding switches changing the V_{DS} to the voltage of the capacitor, which ensures a safe current through the MOSFET.

FIG. 3 is a simplified schematic diagram illustrating a voltage-to-current converter circuit for providing an output current I_{CAL} , that is inversely related to an input voltage that embodies certain aspects of this invention. The voltage-to-current converter circuit 300 in FIG. 3 can be used as the inverse voltage-to-current converter circuit 122 of FIG. 1. As shown in FIG. 3, inverse voltage-to-current converter circuit 300 includes a first terminal 301 and a second terminal 302 for receiving a first input voltage signal and a second input voltage signal from the protected device, e.g., a power MOSFET. Thus, the difference between the first input voltage signal and the second input voltage signal represent a voltage across the first and second terminals. Inverse voltage-to-current converter circuit 300 also include a voltage-to-time converter circuit 310 for providing a time indicator pulse signal T_{CAL} with a pulse width inversely related to a difference between the first and second input voltage signals, a time-to-voltage converter circuit 320 for providing a voltage indicator signal V_{CAL} having a magnitude based on the pulse width of the time indicator pulse signal, a voltage-to-current converter circuit 330 for providing a current indicator signal I_{CAL} , having a magnitude proportional to the voltage indicator signal. The current indicator signal I_{CAL} is configured to have a magnitude inversely related to the difference between the first and the second input voltage signals. The current indicator signal I_{CAL} represents a maximum allowed current based on the pre-set maximum power limit for a voltage across the protective device.

In some embodiments, values for circuit components in the inverse voltage-to-current converter circuit are selected so that the current indicator signal represents the maximum current allowed based on the pre-set maximum power limit for a given voltage across the protective device. As an example, the selection is according to the formula:

$$P_{LMT} = I \Delta V = K_C \frac{R_1 C_1}{R_2 C_2} V_{REF} I_{REF}$$

where:

ΔV is the voltage across the protected device;

K_C is a current sensing ratio;

R_1 , C_1 , and V_{REF} are a resistor, a capacitor, and a voltage reference, respectively, in the voltage-to-time converter circuit;

C_2 and I_{REF} are a capacitor and a current reference, respectively, in the time-to-voltage converter circuit; and

R_2 is a resistor in the voltage-to-current converter circuit.

FIG. 4 is a simplified schematic diagram illustrating a voltage-to-time converter circuit for providing a voltage pulse signal with a pulse width inversely related to a magnitude of an input voltage that embodies certain aspects of this invention. The voltage-to-time converter circuit 400 in FIG. 4 is an example of circuit that can be used as the voltage-to-time conversion circuit 310 of FIG. 3. As shown

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in FIG. 4, the voltage-to-time conversion circuit 400 has two input terminals for coupling to two terminals of a device for monitoring the voltage across the two terminals. In FIG. 4, a first terminal marked 101 (Input) is coupled to an input terminal of the device being monitored, and a second terminal marked 102 (Output) is coupled to an output terminal of the device being monitored. These terminals correspond to the 101 and 102 terminals of the power MOSFET of FIG. 1. A first resistor R_1 is connected to terminal 101 to sense a first current I_P determined by the voltage at terminal 101. Similarly, a second resistor, also having a magnitude R_1 , is connected to terminal 102 to sense a second current I_N determined by the voltage at terminal 102. Current mirror circuits 411 and 412 are used to produce currents I_P and I_N coupled at an internal node 415. A current I_C , which is equal to the difference currents I_P and I_N at node 415, is used to charge a first capacitor C_1 through a first switch SW1. A timer circuit 421 resets and restarts the monitoring cycle periodically, e.g., each period can be 32 μ sec, or other suitable time intervals. Before the start of each cycle, capacitor C_1 is discharged through a second switch SW2. At the start of each period, the output signal T_{CAL} is set to high, and capacitor C_1 starts to be charged by the current I_C from node 415, which is the difference between the currents between I_P and I_N . As the voltage V_1 on C_1 reaches a reference voltage V_{REF} , comparator CMP 422 triggers the D-latch 424 to lower the output signal T_{CAL} . The larger the current $I_C = (I_P - I_N)$, the quicker capacitor C_1 charges up; conversely, the smaller the current $I_C = (I_P - I_N)$, the slower capacitor C_1 charges up. Therefore, the length, or pulse width, of output pulse signal T_{CAL} is inversely related to the magnitude of current $I_C = (I_P - I_N)$. As described, the voltage-to-time conversion circuit 400 of FIG. 4 converts voltage across a power MOSFET to a pulse signal T_{CAL} , which is a time indicator pulse signal with a pulse width related to a charging time of the first capacitor. Further, the maximum pulse width of T_{CAL} is determined by the period of the timer circuit, which can be a selectable parameter of the circuit. In FIG. 4, a second switch SW2 is used to discharge capacitor C_1 in response to the T_{CAL} signal in every cycle.

FIG. 5 is a simplified schematic diagram for a time-to-voltage converter circuit that embodies certain aspects of this invention. As shown in FIG. 5, a time-to-voltage converter circuit 500 is configured for providing an output voltage having a magnitude based on a length of an input voltage pulse signal. Time-to-voltage converter circuit 500 is an example of time-to-voltage converter circuit that can be used as the time-to-voltage converter circuit 320 in FIG. 3.

In FIG. 5, the time-to-voltage converter circuit 500 includes a reference current source for generating a reference current signal I_{REF} , a first capacitor C_1 , and a second capacitor C_2 . The first capacitor C_1 is coupled to the reference current source 501 through a first switch 511 controlled by the time indicator pulse signal T_{CAL} , which is provided by the voltage-to-time converter 400 in FIG. 4. The time-to-voltage converter circuit 500 is configured to charge the capacitor C_1 by the reference current source 501 during an on-time of the time indicator pulse signal T_{CAL} , and to produce a voltage 520 on the capacitor C_1 as the voltage indicator signal V_{CAL} .

The time-to-voltage converter circuit 500 also has a second switch 513 controlled by a control signal Φ_1 , for controlling the transfer of charges from capacitor C_1 to a second capacitor C_2 for holding the voltage at node 520 to provide the voltage indicator signal V_{CAL} . A third switch 515 is controlled by a second control signal Φ_2 . For controlling the discharge of capacitor C_2 in every cycle. The time-to-

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voltage conversion circuit is configured to provide an output signal V_{CAL} , whose magnitude is based on the length of input voltage pulse signal T_{CAL} . The operation of the time-to-voltage conversion circuit in FIG. 3 is described below with reference to FIG. 6.

FIG. 6 is a waveform diagram illustrating the waveforms of T_{CAL} , Φ_1 , and Φ_2 described above in connection to FIG. 5 that embodies certain aspects of this invention. It can be seen in FIG. 5 that when T_{CAL} is on and Φ_1 and Φ_2 are off, capacitor C_2 is charged for a time duration T_{CAL} . Next, T_{CAL} is low, Φ_1 is turned on, and C_3 is charged by the voltage on C_2 . In this circuit, $C_2 \gg C_3$. Because the capacitance of C_3 is much smaller than C_2 , the voltage on C_2 essentially transfers to C_3 with negligible voltage. Then, Φ_2 turns on to discharge C_2 . The output V_{CAL} at capacitor C_3 can be expressed as follows.

$$V_{CAL} = \frac{I_{REF}}{C_2} T_{CAL}$$

In this circuit, $C_3 \ll C_2$. For example, the capacitance of C_2 can be 10 times the capacitance of C_3 . In a specific example, C_2 may have a capacitance of 10 pF, and C_3 may have a capacitance of 1 pF. The output voltage V_{CAL} can be held by a small capacitor C_3 for processing in the next stage.

FIG. 7 is a simplified schematic diagram illustrating a voltage-to-current converter circuit that embodies certain aspects of this invention. The voltage-to-current converter circuit 700 is an example of voltage-to-current converter circuit that can be used as the voltage-to-current converter circuit 330 of FIG. 3. In this example, the voltage-to-current converter circuit 700 a current regulator includes a current regulator for providing a current signal I_{CAL} , having a magnitude proportional to the an input voltage V_{CAL} , which can be an output from the time-to-voltage converter circuit of FIG. 5. The current signal I_{CAL} , has a magnitude inversely related to the input voltage of the inverse voltage-to-current converter circuit 122 illustrated in FIG. 1 and the inverse voltage-to-current converter circuit 300 illustrated in FIG. 3.

In FIG. 7, V_{CAL} , is an input to an operational amplifier 710 at the + input. The current I_{CAL} , is provided by a current mirror 720 and coupled to a resistor R2. Current mirror 720 is coupled to the output of the operational amplifier 710 in a feedback loop. The current I_{CAL} , is determined such that the voltage at the—input of the operation amplifier 710 is equal to V_{CAL} . Therefore,

$$I_{CAL} = \frac{V_{CAL}}{R_2}$$

I_{CAL} , is provided at the output by a current mirror circuit 720. Thus, the voltage-to-current converter circuit 700 produces an output current I_{CAL} , proportional to the voltage indicator signal V_{CAL} .

FIG. 8 is a simplified schematic diagram of a current selector circuit that embodies certain aspects of this invention. The current selector circuit 800 is an example of current limit selector circuit that can be used as the current limit selector circuit 124 of FIG. 1. As shown in FIG. 8, the current selector circuit includes a current comparator circuit ICMP (810) and a multiplexor circuit IMUX (820). The current comparator 810 is used to compare I_{CAL} , to I_{SET} , the current limit I_{SET} can be provided to the circuit either internally or externally on system board. The current multiplexer 820 to

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choose the smaller one between I_{CAL} to I_{SET} . To be the target current limit signal I_{LMT} . The functions of the current selector circuit can be described with the following expressions.

$$\text{If } I_{CAL} < I_{SET}, \text{ set } I_{LMT} = I_{CAL};$$

$$\text{If } I_{CAL} \geq I_{SET}, \text{ set } I_{LMT} = I_{SET};$$

In other words, the lower of the two input signals I_{CAL} and I_{SET} is used as the current limit provided to the over-current protection circuit.

Referring to FIG. 1, in the protection circuit 100, the smaller one between I_{CAL} to I_{SET} is sent to the over-current protection circuit 110 to control the gate of power MOSFET in order to limit current through power MOSFET 20 at a current limit I_{LMT} .

Given a power limit, the values of the components in the over-power protection circuit can be determined to provide a target current limit signal to the over-current protection circuit 110 to control the gate of power MOSFET in order to limit the power to not exceed the power limit. From the equations described above.

$$T_{CAL} = R_1 C_1 \frac{V_{REF}}{\Delta V}$$

$$V_{CAL} = \frac{I_{REF}}{C_2} T_{CAL}$$

$$I_{CAL} = \frac{V_{CAL}}{R_2}$$

where ΔV is the voltage across the protected device.

We can have

$$I_{CAL} = \left(\frac{R_1 C_1}{R_2 C_2} V_{REF} I_{REF} \right) / \Delta V$$

Let the sensing current ratio be defined as K_C ,

$$K_C = \frac{I}{I_{SENSE}}$$

in which I is the current flowing through the power MOSFET.

When the over-current protection (OCP) condition occurs with the current limit set by I_{CAL} , we have the following relationship:

$$P_{LMT} = I_{CAL} \Delta V = K_C \frac{R_1 C_1}{R_2 C_2} V_{REF} I_{REF}$$

where ΔV is the voltage across the protected device.

With a given P_{LMT} , the appropriate values of the parameters can be selected. With these parameters, I_{CAL} can be generated reversely proportional to ΔV . For example, depending on the embodiments, K_C can be between 10^1 to 10^6 , R_1 and R_2 can be 20 K Ω to 800 K Ω , C_1 and C_2 can be 1 pF to 10 pf, V_{REF} can be 1 V to 5V, and I_{REF} can be 1 μ A to 5 μ A, etc.

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In these embodiments, I_{CAL} can be made to be less dependent to variations in process conditions, supply voltage, and operating temperature. For example, R_1 , R_2 , C_2 and C_2 can be designed to match each other, V_{REF} can be determined from a bandgap voltage circuit, and I_{REF} can be derived from a bandgap voltage crossing a zero-Tc resistor.

What is claimed is:

1. An inverse voltage-to-current conversion circuit for providing a current that is inversely related to an input voltage, comprising:

a first input terminal and a second input terminal for receiving the input voltage between the first and the second input terminals;

a voltage-to-time converter circuit for providing a time indicator pulse signal with a pulse width related to inverse magnitude of the input voltage, wherein the voltage-to-time converter circuit comprises a first capacitor and a latch outputting the time indicator pulse signal with the pulse width, the pulse width being a charging time of the first capacitor, and the time indicator pulse signal switching from logic high to logic low when a voltage across the first capacitor reaches a reference voltage;

a time-to-voltage converter circuit for providing a voltage indicator signal having a magnitude based on the pulse width of the time indicator pulse signal; and

a voltage-to-current converter circuit for providing a current indicator signal having a magnitude proportional to the voltage indicator signal, the current indicator signal being inversely related to the magnitude of the input voltage.

2. The inverse voltage-to-current conversion circuit of claim 1, wherein the voltage-to-time converter circuit further comprises:

a first resistor coupled to the first input terminal for sampling a first current; and

a second resistor coupled to the second input terminal for sampling a second current, wherein the first capacitor is configured to be charged by a current related to a difference between the first current and the second current.

3. The inverse voltage-to-current conversion circuit of claim 1, wherein the time indicator pulse signal is in a time domain.

4. The inverse voltage-to-current conversion circuit of claim 1, wherein circuit components are selected so that the current indicator signal represents a maximum current allowed.

5. The inverse voltage-to-current conversion circuit of claim 1, wherein the current indicator signal is based on a pre-set maximum power limit for a voltage across a protective device.

6. The inverse voltage-to-current conversion circuit of claim 1, wherein the time-to-voltage converter circuit comprises:

a reference current source configured to generate a reference current signal; and

a capacitor coupled to the reference current source through a switch controlled by the time indicator pulse signal;

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wherein the time-to-voltage converter circuit is configured to charge the capacitor by the reference current source during an on-time of the time indicator pulse signal; and

to produce a voltage on the capacitor as the voltage indicator signal.

7. The inverse voltage-to-current conversion circuit of claim 1, wherein the voltage-to-current converter circuit comprises a current regulator including an operational amplifier configured to provide an output current proportional to the voltage indicator signal.

8. A method for producing a current that is inversely proportional to a first voltage, comprising:

sensing a first voltage across a first terminal and a second terminal;

forming, by a voltage-to-time converter comprising a first capacitor and a latch, a time indicator pulse signal with a pulse width inversely related to a magnitude of the first voltage, wherein forming the time indicator pulse signal comprises:

charging the first capacitor; and

operating the latch to output the time indicator pulse signal, the pulse width being a charging time of the first capacitor, the time indicator pulse signal switching from logic high to logic low when a voltage across the first capacitor reaches a reference voltage;

forming a second voltage having a magnitude based on a length of the time indicator pulse signal; and

forming a current signal having a magnitude proportional to the second voltage.

9. The method of claim 8, wherein the current signal is configured to have a magnitude inversely related to the first voltage.

10. The method of claim 8, wherein the time indicator pulse signal is in a time domain.

11. The method of claim 8, wherein forming the time indicator pulse signal further comprises:

sampling a first current at a first resistor coupled to the first terminal; and

sampling a second current at a second resistor coupled to the second terminal, wherein the first capacitor is charged by a current related to a difference between the first current and the second current.

12. The method of claim 8, wherein the current signal represents a maximum current allowed.

13. The method of claim 8, wherein the current signal is based on a pre-set maximum power limit for a voltage across a protective device.

14. The method of claim 8, wherein forming the second voltage having a magnitude based on a length of the time indicator pulse signal comprises:

generating a reference current source signal using a reference current source;

charging a capacitor by the reference current source during an on-time of the time indicator pulse signal; and

producing a voltage on the capacitor as the second voltage.

15. The method of claim 14 wherein the capacitor is coupled to the reference current source through a switch controlled by the time indicator pulse signal.

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